Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A balloon of diameter $20\ \mathrm{m}$ weighs $100\ \mathrm{kg}$. What is its pay - load, if it is filled with He at $1\ \mathrm{atm}$ and $27^{\circ}\mathrm{C}$. Density of air is $1.2\ \mathrm{kg\ m^{-3}}$

(A) $424.67 \times 10^{4}\ \mathrm{g}$

(B) $424.67 \times 10^{2}\ \mathrm{g}$

(C) $424.67 \times 10^{5}\ \mathrm{g}$

(D) None of these

Q.2

The behaviour of a real gas is usually depicted by plotting compressibility factor Z versus P at a constant temperature. At high temperature and high pressure, Z is usually more than one. This fact can be explained by van der Waals equation when

(A) the constant ' a ' is negligible and not 'b'

(B) the constant ' b ' is negligible and not 'a'

(C) both constants ' a ' and ' b ' are negligible

(D) Both the constants ' a ' and 'b' are not negligible.

Q.3

50 litre of dry $\mathrm{N_2}$ is passed through $36\ \mathrm{g}$ of $\mathrm{H_2O}$ at $27^{\circ}\mathrm{C}$. After passage of gas, there is a loss of $1.20\ \mathrm{g}$ in water. Calculate vapour pressure of water.

(A) $32.48\ \mathrm{mm}$

(B) $24.95\ \mathrm{mm}$

(C) $30.56\ \mathrm{mm}$

(D) None of these

Q.4

The average speed of an ideal gas molecule at $27^{\circ}\mathrm{C}$ is $0.3\ \mathrm{msec^{-1}}$. Calculate average speed at $927^{\circ}\mathrm{C}$.

(A) $0.6\ \mathrm{msec^{-1}}$

(B) $1.0\ \mathrm{msec^{-1}}$

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(C) $1.4\ \text{msec}^{-1}$

(D) $2.8\ \text{msec}^{-1}$

Q.5

Pure O_2 diffuses through an aperture in
224 seconds, whereas a mixture of O_2 and another gas containing $80\%O_2$ diffuses from the same in 234sec. What is the molecular weight of gas ?

(A) 46.6

(B) 34.92

(C) 48.6

(D) None

Q.6

What is kinetic energy of $1\ \text{g}$ of O_2 at 47°C

(A) $1.24 \times 10^2\ \text{J}$

(B) $2.24 \times 10^2\ \text{J}$

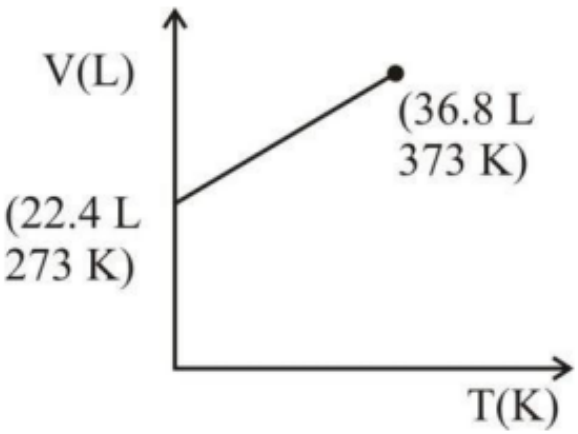
(C) $1.24 \times 10^3\ \text{J}$

(D) $3.24 \times 10^2\ \text{J}$

Q.7

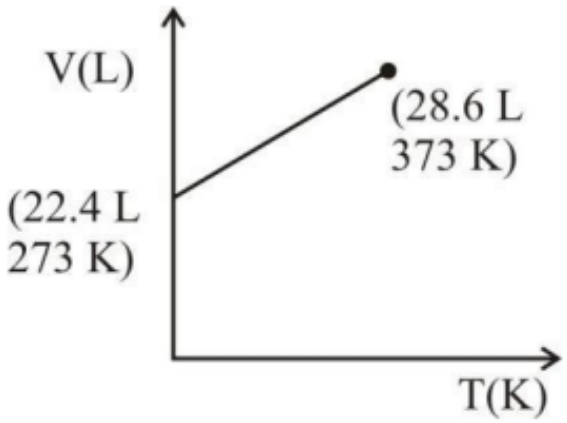
Which of the following volume $(\text{V})-$ temperature (T) plots represents the behavior of one mole of an ideal gas at one atmospheric pressure

(A)

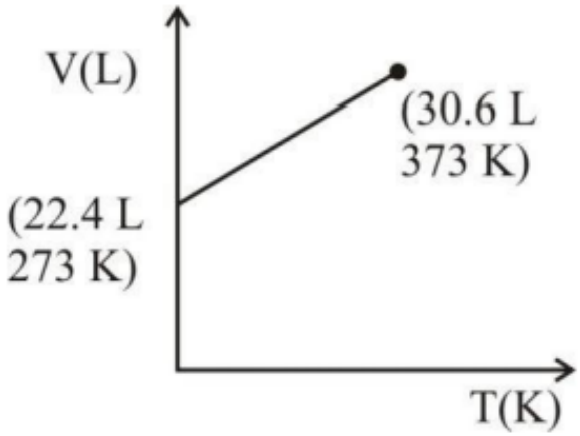


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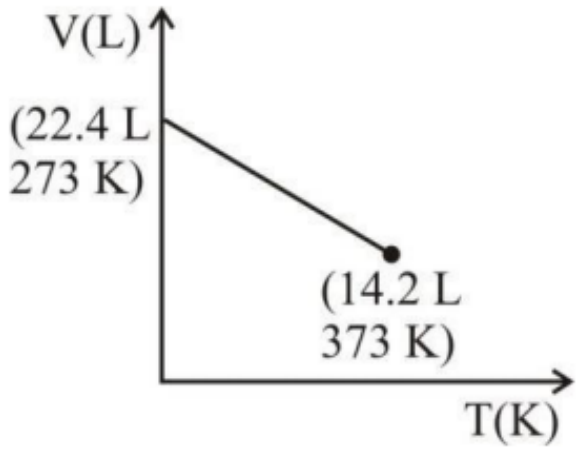
(B)



(C)



(D)



Q.8

The rate of diffusion of SO_2 and O_2 are in the ratio

(A) $1 : \sqrt{2}$

(B) $1 : 32$

(C) $1 : 2$

(D) $1 : 4$

Q.9

The rms speed of hydrogen is $\sqrt{7}$ times the rms speed of nitrogen, if T is the temperature of the gas then

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(A) $T_{H_2} = T_{N_2}$

(B) $T_{H_2} > T_{N_2}$

(C) $T_{H_2} < T_{N_2}$

(D) $T_{H_2} > \sqrt{7} T_{N_2}$

Q.10

The volume X of H_2 gas effuses through a hole in a container in 5 seconds. The time taken for the effusion of the same volume of the specified gas under identical conditions is

(A) 10 seconds: He

(B) 20 seconds : O_2

(C) 25 seconds : CO

(D) 55 seconds : CO_2

Q.11

The ratio of root mean square speed to average speed of a gas molecule at a particular temperature is

(A) $1.086 : 1$

(B) $1 : 1.086$

(C) $2 : 1.086$

(D) $1.086 : 2$

Q.12

Which of the following statements regarding the mean free path of gaseous molecules is correct ?

(A) Directly proportional to the number of molecules per unit volume

(B) Directly proportional to the average speed of molecules

(C) Inversely proportional to the square of molecular diameter

(D) Inversely proportional to the temperature of the gas

Q.13

A balloon has a maximum capacity of $20\ \mathrm{L}$. At one atmospheric pressure $10\ \mathrm{L}$ of air is filled in the balloon. It will burst when pressure is (assuming isothermal condition)

(A) $0.5\ \mathrm{atm}$

(B) $0.4\ \mathrm{atm}$

(C) $0.7\ \mathrm{atm}$

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(D) 0.8 atm

Q.14

If the density of a certain gas at $30^{\circ}\mathrm{C}$ and 768 torr is $1.35\ \mathrm{kg/m^3}$, then density at STP is

(A) $1.48\ \mathrm{kg/m^3}$

(B) $1.27\ \mathrm{kg/m^3}$

(C) $1.35\ \mathrm{kg/m^3}$

(D) $1.00\ \mathrm{kg/m^3}$

Q.15

A $1.00\ \mathrm{L}$ vessel containing $1.00\ \mathrm{g}\mathrm{H_2}$ gas at $27^{\circ}\mathrm{C}$ is connected to a $2.00\ \mathrm{L}$ vessel containing $88.0\ \mathrm{g}\mathrm{CO_2}$ gas, at also $27^{\circ}\mathrm{C}$. When the gases are completely mixed, total pressure is

(A) 20.525 atm

(B) 4.105 atm

(C) 16.420 atm

(D) 730.69 atm

Q.16

Which has the maximum value of the mean free path?

(A) $\mathrm{CO_2}$

(B) $\mathrm{H_2}$

(C) $\mathrm{O_2}$

(D) $\mathrm{N_2}$

Q.17

The vapour density of gas X is 4 times that of Y. If molecular mass of Y is M, then molecular mass of X is

(A) $\mathrm{M}/2$

(B) $4\mathrm{M}$

(C) $\mathrm{M}/4$

(D) $2\mathrm{M}$

Q.18

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Three different gases X, Y, Z of molecular masses $32, 64, 71$ were enclosed in a vessel at constant temperature till equilibrium is reached. Which of the following statements is/are true ?

(A) Gas X will be at the top of the vessel

(B) Gas Y will be at the top of the vessel

(C) Gas X will be at the bottom and Z will be at the top

(D) Gases will form a homogeneous mixture.

Q.19

For Xg of ideal gas V and $\mathrm{V_0}$ are the volumes at $\mathrm{t^\circ C}$ and $0^\circ \mathrm{C}$ respectively at 1 atm pressure so that $\mathrm{V/V_0 = (1 + \alpha t)}$. Here, α is

(A) < 1

(B) < 0

(C) > 1

(D) $= 0$

Q.20

Three gases X, Y, Z are enclosed in a container and have partial pressure as p_1, p_2, p_3 respectively at T Kelvin. The total pressure at T Kelvin will be

(A) always equal to $\mathrm{(p_1 + p_2 + p_3)}$

(B) $< \mathrm{(p_1 + p_2 + p_3)}$

(C) $> (p_1 + p_2 + p_3)$

(D) $= (p_1 + p_2 + p_3)$ only if the gases do not react chemically at T.

Q.21

A sample of gas occupies 100 litres at 1 atm pressure and at $0^\circ \mathrm{C}$. If the volume of the gas is to be reduced to 5 litres at the same temperature, what additional pressure(in atm) must be applied ?

Q.22

At an underwater depth of $250\mathrm{ft}$; the pressure is $8.38\ \mathrm{atm}$. What should be the value of 2 times of mole percent(\%) of oxygen in the diving gas for the partial pressure of oxygen in the mixture to be $0.21\ \mathrm{atm}$, the same as it is in air at $1\ \mathrm{atm}$?

Q.23

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1 mole of SO_2 occupies a volume of $350ml$ at $300\ K$ and $50\ atm$ pressure. The compressibility factor of the gas is Z, then what is the value of $Z \times 10$ (round off to the nearest integer)?

Q.24

A cylindrical balloon of $21\ cm$ diameter is to be filled up with H_2 at NTP from a cylinder containing the gas at $20\ atm$ at $27^\circ C$. The cylinder can hold 2.82 litre of water at NTP. How many balloons can be filled up.

Q.25

Equal volumes of two gases which do not react together are enclosed in separate vessels. Their pressures are $100\ mm$ and $400\ mm$ respectively. If the two vessels are joined together, then what will be the pressure (in mm) of the resulting mixture (temperature remaining constant)

Q.26

A gas of volume $100cc$ is kept in a vessel at pressure 10^4 Pa maintained at temperature $24^\circ C$. If now the pressure is increased to
$10^5\ Pa$, keeping the temperature constant, then the volume of the gas(in cc) becomes

Q.27

$16\ g$ of oxygen and $3\ g$ of hydrogen are mixed and kept at $760\ mm$ pressure and $0^\circ C$. The total volume (in ml) occupied by the mixture will be nearly

Q.28

A weather balloon filled with hydrogen at $1\ atm$ and $27^\circ C$ has volume equal to 12000 litres. On ascending it reaches a place where the temperature is $-23^\circ C$ and pressure is $0.5\ atm$. The volume of the balloon in litres is

Q.29

Three grams of helium diffuses from a container in $15\ min$. The mass(in g) of sulphur dioxide diffusing from the same container over the same time interval will be

Q.30



In the above experiment, the value of P in mm is

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Q.1 Answer:

424.67×10^4 g

Solution:

Weight of balloon $= 100$ kg $= 10 \times 10^4$ g

Volume of balloon

$= \frac{4}{3}\pi r^3 = \frac{4}{3} \times \frac{22}{7} \times \left(\frac{20}{2} \times 100\right)^3$

$= 4190 \times 10^3$ L

Weight of gas (He) **in balloon**

$= \frac{\mathrm{PV_m}}{\mathrm{RT}} = \frac{1 \times 4190 \times 10^3 \times 4}{0.082 \times 300} = 68.13 \times 10^4$ g

∴ Total weight of gas and balloon

$= 68.13 \times 10^4 + 10 \times 10^4 = 78.13 \times 10^4$ g

Weight of air displaced

$= \frac{1.2 \times 4190 \times 10^6}{10^3} = 502.8 \times 10^4$ g

$\text{Payload} = \text{weight of air displaced} - (\text{weight of balloon} + \text{weight of gas})$

Payload

$= 502.8 \times 10^4 - 78.13 \times 10^4$

$= 424.67 \times 10^4$ g

Q.2 Answer:

the constant ' a ' is negligible and not 'b'

Solution:

$\left(\mathrm{P} + \frac{\mathrm{n^2 a}}{\mathrm{V^2}}\right)(\mathrm{V} - \mathrm{nb}) = \mathrm{nRT}$

At low pressures, 'b' can be ignored as the volume of the gas is very high. At high temperatures 'a' can be ignored as the pressure of the gas is high.

$\therefore P(V - b) = RT$

$PV - Pb = RT \Rightarrow PV = RT + Pb$

$\frac{PV}{RT} = Z = 1 + \frac{Pb}{RT}$

Hence, (A) is correct.

Q.3 Answer:

24.95 mm

Solution:

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The water vapours occupies the volume of N_2 gas i.e. 50 litre

$\therefore$ For H_2O vapour V = 50 litre, w = 1.20 g

T = 300 K, m = 18

$PV = \frac{W}{m}RT$

$\therefore$ or$P \times 50 = \frac{1.2}{18} \times 0.0821 \times 300$

$\therefore P = 0.03284$ atm

$= 24.95$ mm

Q.4 Answer:

$0.6\ msec^{-1}$

Solution:

$u_{av} = \sqrt{\frac{8RT}{\pi M}}$

Given $u_{av} = 0.3\ msec^{-1}$ **at** 300 K

$u_1 = 0.3 = \sqrt{\frac{8R \times 300}{\pi M}}$

at $T = 273 + 927 = 1200$ K

$u_2 = \sqrt{\frac{8R \times 1200}{\pi M}}$

$\frac{u_2}{0.3} = \sqrt{\frac{1200}{300}} u_2 = 0.6\ msec^{-1}$

Q.5 Answer:

46.6

Solution:

For gaseous mixture $80\% O_2, 20\%$ **gas Average molecular weight** M_m

$= \frac{32 \times 80 + 20 \times m}{100}$

Now, for diffusion of gaseous mixture and pure O_2

$\frac{r_{O_2}}{r_m} = \sqrt{\frac{M_m}{M_{O_2}}}$ or $\frac{V_{O_2}}{E_{O_2}} \times \frac{E_m}{V_m} = \sqrt{\frac{M_m}{M_{O_2}}}$

$\therefore \quad \frac{1}{224} \times \frac{234}{1} = \sqrt{\frac{M_m}{32}} \therefore M_m = 34.92$

$\therefore \frac{32 \times 80 + 20 \times m}{100} = 34.92$

$\therefore m = 46.6$

Q.6 Answer:

1.24×10^2 J

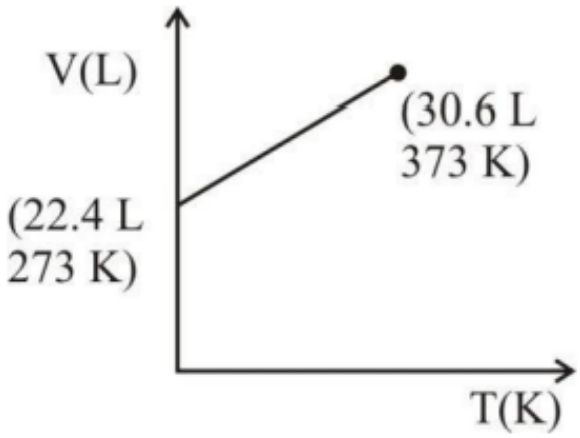
Solution:

K.E $= \frac{3}{2}nRT = \frac{3}{2} \times \frac{1}{32} \times 8.314 \times 320^\circ K.$

$= 1.24 \times 10^2$ J

Q.7 Answer:

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Solution:

$\frac{V_1}{V_2} = \frac{T_1}{T_2} \therefore V_2 = V_1 \cdot \frac{T_2}{T_1} = \frac{22.4 \times 373}{273} = 30.6\ L$

Q.8 Answer:

$1 : \sqrt{2}$

Solution:

$10^4 \cdot 100 = 10^5 \times V_2$

$V_2 = 10cc214$

$\frac{r_{SO_2}}{r_{O_2}} = \sqrt{\frac{M_{O_2}}{M_{SO_2}}} = \sqrt{\frac{32}{64}} = \frac{1}{\sqrt{2}}$

Q.9 Answer:

$T_{H_2} < T_{N_2}$

Solution:

$U_{rms}\,(H_2) = \sqrt{7} U_{rms}\,(N_2)$

or $\sqrt{\frac{2r\ T_{H_2}}{2}} = \sqrt{7} \times \sqrt{\frac{3RT_{N_2}}{28}}$

$\therefore T_{N_2} = 2\ T_{H_2}$ **or**

$T_{N_2} > T_{H_2}$

Q.10 Answer:

20 seconds : O_2

Solution:

$\frac{t_2}{t_1} = \left(\frac{M_2}{M_1}\right)^{1/2}$

$t_2/t_1 = \sqrt{2}$**, for the choice b,** $t_2/t_1 = 4$**. For the choice c,** $t_2/t_1 = \sqrt{14}$**. For the choice d,** $t_2/t_1 = \sqrt{22}$**. Thus, the choice** b **is correct.**

Q.11 Answer:

$1.086 : 1$

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Solution:

$c_{rms} = \sqrt{3RT/M}$ **and** $c_{av} = \sqrt{8RT/\pi M}$

Hence $\frac{c_{rms}}{c_{av}} = \sqrt{\frac{3\pi}{8}} = \sqrt{\frac{3\times 3.14}{8}} = \sqrt{\frac{9.42}{8}}$

$= \sqrt{1.165} = 1.086$

Q.12 Answer:

Inversely proportional to the square of molecular diameter

Solution:

The expression of mean free path is

$\lambda = \frac{1}{\sqrt{2}\pi N^* \sigma^2}$

Where N^* **is the number of molecules per unit volume and** σ **is the molecular diameter.**

Q.13 Answer:

0.4 atm

Solution:

The balloon would burst when $\mathrm{V} > 20$ L

$\mathrm{P}_1\ \mathrm{V}_1 = \mathrm{P}_2\ \mathrm{V}_2$

$1 \times 10 = \mathrm{P}_2 \times 20$

$\mathrm{P}_2 = 0.5$ atm (no bursting)

Thus, a pressure below 0.5 **atm, it would burst thus, (b)**

Q.14 Answer:

$1.48\ \mathrm{kg/m^3}$

Solution:

$PV = nRT = \frac{w}{m}RT$

$\therefore \quad P = \frac{w}{Vm}RT$

$P = \frac{dRT}{m}$

$\therefore \quad d = \frac{Pm}{RT}$

$\frac{d_2}{d_1} = \frac{P_2}{T_2}\frac{T_1}{P_1}$

$d_2(\mathrm{STP}) = \frac{1.35}{273} \times \frac{760\times 303}{768}$

$= 1.48\ \mathrm{kg/m^3}$

Q.15 Answer:

20.525 atm

Solution:

$P_{\text{Total}} = \left(\frac{n_1+n_2}{V_1+V_2}\right)RT$

$= \frac{2.5}{3} \times 0.0821 \times 300$ atm

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$= 20.525$ atm

Q.16 Answer:

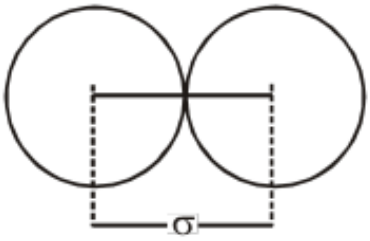
H_2

Solution:

Mean free path

$\lambda = \frac{1}{\sqrt{2}\pi\sigma^2 N}$

$\lambda \propto \frac{1}{\sigma^2}$



σ is collision diameter thus, smaller the value of σ larger the value of λ

Q.17 Answer:

4M

Solution:

Mol. mass of $Y = M$ $\quad \therefore$ **V.D. of** $Y = \frac{M}{2}$

V.D. of $X = 4 \times \frac{M}{2} = 2M$

$\therefore$ **Mol mass of** $X = 2 \times 2M = 4M$.

Q.18 Answer:

Gases will form a homogeneous mixture.

Solution:

Free intermixing of gases results in uniform mixture.

Intermixing of the particles of substances on their own is called diffusion. A motion of the substances relative to one another throughout a definite region of space in which mixing is taking place. This relative motion is called diffusion. One of the properties of gases is that they mix with each other. When they do so, they become a solution-a homogeneous mixture. Each component of the mixture shares the same temperature and volume. (Remember that gases expand to fill the volume of their container; gases in a mixture do that as well.)

Q.19 Answer:

< 1

Solution:

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According to Charle's law $\alpha = \frac{1}{273}$.

Q.20 Answer:

$= (p_1 + p_2 + p_3)$ only if the gases do not react chemically at T.

Solution:

This is in accordance with Dalton's law.

Dalton's Law states that the pressure exerted by a mixture of gases in a fixed volume is equal to the sum of the pressures that would be exerted by each gas alone in the same volume.

Thus, $\mathrm{P} = \mathrm{P}_1 + \mathrm{P}_2 + \mathrm{P}_3$

Q.21 Answer:

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Solution:

$\mathrm{Here\ P_1 = 1\ atm} \quad \mathrm{P_2 = ?}$

$\mathrm{V_1 = 100\ litre} \quad \mathrm{V_2 = 5\ litre}$

$\mathrm{T_1 = 273\ K} \quad \mathrm{T_2 = 273\ K}$

As temperature is constant

Then from Boyle's law

$\mathrm{P_1\ V_1 = P_2\ V_2}$

or $1 \times 100 = \mathrm{P_2} \times 5$

$\therefore \mathrm{P_2} = 20\ \mathrm{atm}$

$\therefore$ **Additional pressure that should be applied** $= \mathrm{P_2 - P_1} = 20 - 1 = 19\ \mathrm{atm}$

Q.22 Answer:

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Solution:

Since $\mathrm{P_{O_2} = X_{O_2} \times P_{total}}$ **then** $\mathrm{X_{O_2}} = \frac{\mathrm{P_{O_2}}}{\mathrm{P_{total}}}$

$X_{\mathrm{O_2}} = \frac{0.21}{8.38} = 0.025$

So, the diving gas should contain 2.5% **of** $\mathrm{O_2}$

Q.23 Answer:

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Solution:

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$P = 50$ atm $\quad V = 350\text{ml} = 0.350$ litre
$n = 1$ mole
$T = 300$ K$_u$**A**$Z = \frac{PV}{nRT}$
$\therefore Z = \frac{50 \times 0.350}{1 \times 0.082 \times 300} = 0.711$
Thus SO_2 is more compressible than expected from ideal behavior

Q.24 Answer:

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Solution:

Volume of 1 balloon which has to be filled
$= \frac{4}{3}\pi\left(\frac{21}{2}\right)^3 = 4851\text{ml}$
$= 4.851\text{litre}$
Let n balloons be filled, then volume of H_2 occupied by balloons $= 4.851 \times n$
Also, the cylinder will not be empty and it will occupy a volume of $H_2 = 2.82$ litre.
$\therefore$ Total volume occupied by H_2 at NTP $= 4.851 \times n + 2.82$ **litre**
$\therefore$ At STP

$P_2 = 1$ atm	Available H_2
$V_1 = 4.851 \times n + 2.82$	$P_2 = 20$ atm
$T_1 = 273$ K	$T_2 = 300$ K
$\frac{P_1 V_1}{T_2} = \frac{P_2 V_2}{T_2}$	
$\frac{1 \times (4.85\ln + 2.82)}{273}$	$V_2 = 2.82$ litre
or	$= \frac{20 \times 2.82}{300}$ $\therefore n = 10$

Q.25 Answer:

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Solution:

When two vessels are joined together, the volume will be doubled hence effective pressure will be halved
$P = \frac{P_1 + P_2}{2} = \frac{100 + 400}{2} = 250$

Q.26 Answer:

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Solution:

$P_1\ V_1 = P_2\ V_2$ **at constant** T
$10^4 \cdot 100 = 10^5 \times V_2\ V_2 = 10\text{cc}$

Q.27 Answer:

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Solution:

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n of $O_2 = \frac{16}{32} = \frac{1}{2}$
n of $H_2 = \frac{3}{2}$
Total no. of moles $= \frac{3}{2} + \frac{1}{2} = 2$
$V = \frac{nRT}{P} = \frac{2 \times .082 \times 273}{1} = 44.8\text{lit}$
$= 44800\text{ml}$

Q.28 Answer:

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Solution:

$V_2 = \frac{P_1}{P_2}\frac{T_2}{T_1} \cdot V_1 = \frac{1}{0.5} \times \frac{250}{300} \times 12000$ lit
$= 20000$ lit

Q.29 Answer:

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Solution:

We have $r_2 = \left(\frac{M_1}{M_2}\right)^{1/2} r_1 = \left(\frac{4}{64}\right)^{1/2} r_1 = \frac{r_1}{4}$
Since $r = n/t$, we get $r_2/r_1 = n_2/n_1$
i.e. $n_2 = (r_2/r_1)\, n_1$
$n_2 = (1/4)\left(3g/4g\ \text{mol}^{-1}\right) = \frac{3}{16}\ \text{mol}$
$m_2 = \left(\frac{3}{16}\ \text{mol}\right)\left(64\ \text{g mol}^{-1}\right) = 12\ \text{g}$

Q.30 Answer:

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Solution:

After removal of the partition, the partial pressure of each gas would be halved as volume will be doubled. Hence, the total pressure would remain $125 + 125 = 250$ mm.

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